Minor-I

Design and Analysis of Algorithms

Max marks: 25 | **Max Time: 90 minutes**

Instructions: All questions are compulsory.

1. What are the stable algorithms? Is it necessary for the count sort to be stable? Why? Explain with an example. 3

2. Why is the worst-case running time for bucket sort $O(n^2)$? What changes would you make to the algorithm so that its worst-case running time becomes O(nl gn)'? 3

3 Let P be an array containing n integers. Let t be the lowest upper bound on the number of comparisons of the array elements required to find the minimum and maximum values in an arbitrary array of n elements. What should be the value of t in terms of n. Also, write the algorithm to verify your claim. 4

4. Consider the scenario where the students submit their answer scripts individually to the teacher. The teacher always maintains the collected answer sheets in their order. Suggest the algorithm that the teacher can use. Use this algorithm to arrange the following five answer scripts that are numbered as follows and show all the steps 4
   23, 14, 56, 2, 10
   Compute the number of comparisons that the teacher must have performed. Also, discuss the time complexity.

5. The BFS algorithm has produced the shortest paths from a node to all other nodes in graph G. Can Dijkstra's algorithm be used instead of BFS? In a different scenario, Dijkstra's algorithm has produced the shortest paths from node s to all other nodes in graph C'. Can BFS be used in place of the Dijkstra's algorithm? Explain your answers for both scenarios. 5

6. Let G(V,E) be a directed graph, where V={1,2,3,4,5} is the set of vertices and E is the directed edges, as defined by the following adjacency matrix A. 6
   $$A[i][j] = \begin{cases} 1, & 1 \le j \le i \le 5 \\ 0 & otherwise \end{cases}$$
   A[i][j]= 1 indicates a directed edge from node i to node j. A directed spanning tree of G, rooted at r∈V, is defined as a subgraph T of G such that the undirected version of T is a tree, and T contains a directed path from r to every other vertex in V. Count the number of such directed spanning tree's rooted at vertex 5.

***** All the Best *****

Enrolment No.________________

CS-409: Design and Analysis of Algorithms

Semester: II

May 2025

Max Marks: 60 | Max Time: 3 Hours

Instructions:

- **All questions are compulsory.**
- **Parts of the question answers should be together.**

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1 a. Compute the Prefix function for the following pattern using the KMP string matching algorithm. 2

$$abefabefgabefg$$

b. Give a recurrence relation for the recursive binary search algorithm and derive its average case complexity. 4

c. Solve the recurrence relation $T(n) = 2T\left(n^{\frac{1}{2}}\right) + 1$. Your solution should be asymptotically tight. Specify the base condition you chose for solving this recurrence relation. 4

2 a. Discuss the Greedy approach to solving a problem. Why does this approach fail to produce an optimal solution? List the similarities and differences between the Greedy algorithm and Dynamic Programming. 5

b. What is a state space tree? What are the different ways of searching for an answer node in a state space tree? Explain with an example. 5

3 a. Assuming problem P is to be solved by the Divide-and-Conquer paradigm, what would be your approach regarding the three essential steps you would focus on? List the three criteria that mainly affect the running time of a Divide-and-Conquer algorithm. 5

b. Consider (QA) the Quick sort algorithm to sort integers in non-decreasing order using the last element as pivot. C1 and C2 are the numbers of comparisons made by QA for the given inputs {12, 34, 45, 60, 72, 80} and {40, 10, 50, 30, 60, 20}, respectively. What will the values of C1 and C2 be? (As discussed in class) 5

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4	a.	Compare the approaches of Memoization and Tabulation using a suitable example.	5
	b.	Consider a bucket sort. Why is it not a comparison-based sorting? Give an input instance for comparison-based sorting. Also, compute the time complexity in the best and worst cases.	5
5	a.	Design an algorithm to multiply two numbers using the divide-and-conquer method using a suitable example. (as discussed in class)	5
	b.	How can the 8-Queens problem be solved using backtracking? Explain this with an example.	5
6	a.	Design a bitonic merging network for n = 16, that would sort the bitonic sequence 3, 5, 8, 9, 10, 12, 14, 20, 95, 90, 60, 40, 35, 23, 18, 0, illustrating the outcome of each column of comparators.	5
	b.	Write pseudocode for performing an even-odd transposition sort on a linear array of *n* processors. What is the overall time complexity? Illustrate the sorting process for the following sequence of 8 numbers: 3, 1, 9, 7, 5, 2, 0, 6	5

*************************All the Best*************************